

The Real-Symmetric Polydisc Algebra Is Projective Free

Full-solution packet for arXiv:1103.0899

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Status: candidate full solution, likely valid; human review requested. For every $n \geq 1$, the real-symmetric polydisc algebra A_r is projective free. Radial dilation moves every idempotent matrix over A_r into the smoother projective-free algebra $\partial^{-1}A_r$ while changing it arbitrarily little. An explicit nearby-idempotent similarity transfers the smoother diagonalization back to the original idempotent.

1 The source question

Let $A = A(\overline{\mathbb{D}}^n)$ be the polydisc algebra: the continuous functions on $\overline{\mathbb{D}}^n$ which are holomorphic on $\mathbb{D}^n$. Its real-symmetric real form is

$$A_r = \{f \in A : f(z) = \overline{f(\bar{z})}\}.$$

A commutative unital ring is *projective free* when every finitely generated projective module is free; equivalently, every idempotent matrix over the ring is similar over that ring to $\text{diag}(I_\ell, 0)$.

Sasane [1] proves in Theorem 1.3 that, for every positive integer N , the smoother real-symmetric algebra $\partial^{-N}A_r$ is projective free. At the end of the paper he asks whether A_r itself is projective free.

Then $(\overline{V_0(z)})^{-1} = (e^{-iY/2})^{-1} = e^{iY/2}$. Consequently,

$$V_0(z)(\overline{V_0(z)})^{-1} = e^{iY/2} \cdot e^{iY/2} = e^{iY} = U(0),$$

as required. □

In light of our result above, it is natural to ask the following question: Is the real symmetric polydisc algebra $A_r := \{f \in A : f(z) = \overline{f(\bar{z})} \ (z \in \mathbb{D}^n)\}$ projective free?

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REFERENCES

- [1] J.A. Ball, L. Rodman and I.M. Spitkovsky. Toeplitz corona problem for algebras of almost periodic functions. In *Toeplitz Matrices and Singular Integral Equations* (Pobershau, 2001), 25-37, Operator Theory: Advances and Applications, volume

Figure 1: The source question (PDF page 9).

2 Proof intuition

The missing boundary derivatives can be created without perturbing the idempotent equation. If $P(z)^2 = P(z)$, then $P_r(z) := P(rz)$ is still idempotent. For $r < 1$, it is holomorphic on a larger polydisc and therefore belongs to every smoother algebra in the source theorem. Meanwhile $P_r \rightarrow P$ uniformly.

Idempotents that are sufficiently close in a Banach algebra are similar. Here no abstract stability theorem is needed: the matrix $W = QP + (I - Q)(I - P)$ satisfies $WP = QW$, and it approaches the identity as $Q \rightarrow P$. Taking $Q = P_r$ transports the source theorem's diagonalization of P_r to P .

3 The nearby-idempotent mechanism

Lemma 1. *Let P, Q be idempotents in a unital Banach algebra. If*

$$\|Q - P\| \|2P - I\| < 1, \quad (1)$$

then P and Q are similar.

Proof. Set

$$W = QP + (I - Q)(I - P).$$

Using only $P^2 = P$ and $Q^2 = Q$, one obtains

$$WP = QP = QW. \quad (2)$$

Also

$$W - I = 2QP - Q - P = (Q - P)(2P - I). \quad (3)$$

Condition (1) and the Neumann lemma make W invertible. Equation (2) then gives $Q = WPW^{-1}$. $\square$

4 Full solution

Theorem 2. *For every positive integer n , the real-symmetric polydisc algebra $A_r(\overline{\mathbb{D}}^n)$ is projective free.*

Proof. Let $P \in M_k(A_r)$ be idempotent. For $0 < r < 1$, define

$$P_r(z) = P(rz), \quad z \in \overline{\mathbb{D}}^n. \quad (4)$$

The algebraic identity is preserved pointwise, so $P_r^2 = P_r$. Since r is real and P is real symmetric,

$$P_r(z) = P(rz) = \overline{P(r\bar{z})} = \overline{P_r(\bar{z})}. \quad (5)$$

Moreover, the same formula extends P_r holomorphically to the open polydisc of radius $1/r > 1$. All of its complex partial derivatives are therefore continuous on $\overline{\mathbb{D}}^n$. In particular,

$$P_r \in M_k(\partial^{-1}A_r). \quad (6)$$

Because P is uniformly continuous on the compact closed polydisc,

$$\|P_r - P\|_\infty \longrightarrow 0 \quad (r \uparrow 1). \quad (7)$$

Choose $r < 1$ close enough that $\|P_r - P\| \|2P - I\| < 1$. Lemma 1, applied in the real Banach algebra $M_k(A_r)$, gives an invertible $W \in M_k(A_r)$ such that

$$P_r = W P W^{-1}. \quad (8)$$

By Sasane's Theorem 1.3 for $N = 1$, there are $S_r \in GL_k(\partial^{-1}A_r)$ and an integer ℓ such that

$$S_r^{-1} P_r S_r = \begin{bmatrix} I_\ell & 0 \\ 0 & 0 \end{bmatrix}. \quad (9)$$

Regard S_r as an invertible matrix over A_r and set $S = W^{-1} S_r$. Equations (8)–(9) yield

$$S^{-1} P S = S_r^{-1} W P W^{-1} S_r = S_r^{-1} P_r S_r = \begin{bmatrix} I_\ell & 0 \\ 0 & 0 \end{bmatrix}. \quad (10)$$

Thus every idempotent matrix over A_r has the required constant diagonal form, proving projective freeness. $\square$

5 Verification, scope, and novelty

The proof answers the source question affirmatively for all dimensions. It uses the source's proved Theorem 1.3, not an unproved extension of it. The only transfer ingredients are radial dilation, uniform continuity, and the identities (2)–(3), all valid in the real Banach algebra itself.

The main review points are: (i) formula (4) extends holomorphically to a strictly larger polydisc, which establishes (6); (ii) convergence (7) is in the matrix norm used for the Neumann lemma; and (iii) the order of factors in (8)–(10) is correct. No numerical or computer-assisted claim is used.

On 12 August 2026, the four run indexes and bounded exact/current searches for the arXiv identifier, exact source question, title, author, 'real symmetric polydisc algebra', and 'projective free' found no later explicit resolution. Novelty confidence is moderate-high.

Human-review recommendation. Review as a full solution. The argument is short and reduces the open case directly to the paper's own smoother-algebra theorem through an explicit similarity.

References

- [1] A. Sasane, *Projective Freeness of Algebras of Real Symmetric Functions*, arXiv:1103.0899 (2011), Theorem 1.3 and the question on PDF page 9.